

C++ Solutions — 1D & 2D Array Practice Questions

=== 1D ARRAY SOLUTIONS ===

1) Sum of Array Elements

```
#include <iostream>
using namespace std;
int main(){
    int N; cin>>N;
    long long sum=0,x;
    for(int i=0;i<N;i++){ cin>>x; sum+=x; }
    cout<<sum;
}
```

2) Maximum Element in Array

```
#include <iostream>
using namespace std;
int main(){
    int N; cin>>N;
    long long x,mx=-1e18;
    for(int i=0;i<N;i++){ cin>>x; if(x>mx) mx=x; }
    cout<<mx;
}
```

3) Minimum Element in Array

```
#include <iostream>
using namespace std;
int main(){
    int N; cin>>N;
    long long x,mn=1e18;
    for(int i=0;i<N;i++){ cin>>x; if(x<mn) mn=x; }
    cout<<mn;
}
```

4) Count Even and Odd Numbers

```
#include <iostream>
using namespace std;
int main(){
    int N; cin>>N; long long x;
    int e=0,o=0;
    for(int i=0;i<N;i++){ cin>>x; if(x%2==0) e++; else o++; }
    cout<<e<<" "<<o;
}
```

```
}
```

5) Reverse the Array

```
#include <iostream>
#include <vector>
using namespace std;
int main(){
    int N; cin>>N;
    vector<long long>a(N);
    for(int i=0;i<N;i++) cin>>a[i];
    for(int i=N-1;i>=0;i--){
        cout<<a[i];
        if(i>0) cout<<" ";
    }
}
```

6) Search an Element

```
#include <iostream>
using namespace std;
int main(){
    int N; cin>>N;
    long long x,t; bool f=false;
    long long a;
    for(int i=0;i<N;i++){ cin>>a; if(!f && a==t) f=true; }
    cin>>t;
    cout<<(f?"Found":"Not Found");
}
```

7) Copy Elements to Another Array

```
#include <iostream>
#include <vector>
using namespace std;
int main(){
    int N; cin>>N;
    vector<long long>a(N),b(N);
    for(int i=0;i<N;i++){ cin>>a[i]; b[i]=a[i]; }
    for(int i=0;i<N;i++){
        cout<<b[i];
        if(i<N-1) cout<<" ";
    }
}
```

8) Find Frequency of Each Element

```
#include <iostream>
#include <map>
using namespace std;
```

```

int main(){
    int N; cin>>N; long long x;
    map<long long,int>mp;
    for(int i=0;i<N;i++){ cin>>x; mp[x]++; }
    for(auto &p:mp) cout<<p.first<<"→"<<p.second<<"\n";
}

```

9) Sort Array in Ascending Order

```

#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main(){
    int N; cin>>N;
    vector<long long>a(N);
    for(int i=0;i<N;i++) cin>>a[i];
    sort(a.begin(),a.end());
    for(int i=0;i<N;i++){
        cout<<a[i];
        if(i<N-1) cout<<" ";
    }
}

```

10) Second Largest Element

```

#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main(){
    int N; cin>>N;
    vector<long long>a(N);
    for(int i=0;i<N;i++) cin>>a[i];
    sort(a.begin(),a.end());
    cout<<a[N-2];
}

```

=== 2D ARRAY SOLUTIONS ===

1) Display 3x3 Matrix

```

#include <iostream>
using namespace std;
int main(){
    int a[3][3];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>a[i][j];
    for(int i=0;i<3;i++){

```

```

        for(int j=0;j<3;j++){
            cout<<a[i][j];
            if(j<2) cout<<" ";
        }
        if(i<2) cout<<"\n";
    }
}

```

2) Sum of Matrix Elements

```

#include <iostream>
using namespace std;
int main(){
    int a[3][3], sum=0;
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++){ cin>>a[i][j]; sum+=a[i][j]; }
    cout<<sum;
}

```

3) Row and Column Sum

```

#include <iostream>
using namespace std;
int main(){
    int a[3][3];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>a[i][j];
    for(int i=0;i<3;i++){
        int r=0;
        for(int j=0;j<3;j++) r+=a[i][j];
        cout<<"Row "<<i+1<<" = "<<r<<"\n";
    }
    for(int j=0;j<3;j++){
        int c=0;
        for(int i=0;i<3;i++) c+=a[i][j];
        cout<<"Column "<<j+1<<" = "<<c<<"\n";
    }
}

```

4) Largest & Smallest

```

#include <iostream>
using namespace std;
int main(){
    int a[3][3],mx=-1e9,mn=1e9;
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++){
            cin>>a[i][j];
            if(a[i][j]>mx) mx=a[i][j];
            if(a[i][j]<mn) mn=a[i][j];
        }
}

```

```

        cout<<"Largest = "<<mx<<"\nSmallest = "<<mn;
    }

```

5) Matrix Addition

```

#include <iostream>
using namespace std;
int main(){
    int A[3][3],B[3][3],C[3][3];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>A[i][j];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>B[i][j];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) C[i][j]=A[i][j]+B[i][j];
    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            cout<<C[i][j];
            if(j<2) cout<<" ";
        }
        if(i<2) cout<<"\n";
    }
}

```

6) Matrix Subtraction

```

#include <iostream>
using namespace std;
int main(){
    int A[3][3],B[3][3],C[3][3];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>A[i][j];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>B[i][j];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) C[i][j]=A[i][j]-B[i][j];
    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            cout<<C[i][j];
            if(j<2) cout<<" ";
        }
        if(i<2) cout<<"\n";
    }
}

```

7) Matrix Multiplication

```

#include <iostream>
using namespace std;
int main(){
    int A[3][3],B[3][3],C[3][3]={0};

```

```

    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>A[i][j];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>B[i][j];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            for(int k=0;k<3;k++)
                C[i][j]+=A[i][k]*B[k][j];
    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            cout<<C[i][j];
            if(j<2) cout<<" ";
        }
        if(i<2) cout<<"\n";
    }
}

```

8) Transpose Matrix

```

#include <iostream>
using namespace std;
int main(){
    int A[3][3],T[3][3];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>A[i][j];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) T[j][i]=A[i][j];
    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            cout<<T[i][j];
            if(j<2) cout<<" ";
        }
        if(i<2) cout<<"\n";
    }
}

```

9) Symmetric Matrix Check

```

#include <iostream>
using namespace std;
int main(){
    int A[3][3];
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++) cin>>A[i][j];
    bool f=true;
    for(int i=0;i<3;i++)
        for(int j=0;j<3;j++)
            if(A[i][j]!=A[j][i]) f=false;
    cout<<(f?"Matrix is Symmetric":"Matrix is Not Symmetric");
}

```

10) Sum of Diagonal Elements

```
#include <iostream>
using namespace std;
int main(){
    int A[3][3],sum=0;
    for(int i=0;i<3;i++){
        for(int j=0;j<3;j++){
            cin>>A[i][j];
            if(i==j) sum+=A[i][j];
        }
    }
    cout<<sum;
}
```